Software Requirements Specification

for

Mentor Caring System

Version 1.3 approved

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B09

2026.04.3

Contents

Revision History

Name	Date	Reason For Changes	Version
B09	2026.03.12	Initial approved version	1.0
B09	2026.03.20	Updated to align with Version 1.1 long description and supervisor feedback	1.1
B09	2026.03.28	Updated the scenario, the state transition diagram and user interface based on supervisor feedback and use cases	1.2
B09	2026.04.3	Updated the appendix B with class diagram and sequence diagram	1.3

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the **Mentor Caring System (MCS)**. The purpose of this Software Requirements Specification (SRS) is to describe the systems functions, external interfaces, constraints, and overall behavior.

This document is intended for project stakeholders including developers, project managers, instructors, and future maintainers of the system. It serves as a reference for system design, implementation, and testing.

1.2 Document Conventions

This document follows standard Software Requirements Specification conventions. Section titles and numbering follow the structure defined in the SRS template. Important terms and system components are written with consistent naming throughout the document.

Diagrams such as the context model and use case diagrams are included to visually represent system structure and user interactions.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following readers:

1. **Developers**, who will design and implement the Mentor Caring System;
2. **Project team members**, who need a shared understanding of the system scope and requirements;
3. **Instructors and evaluators**, who will review whether the requirements are complete, clear, and consistent with the project description;
4. **Future maintainers**, who may extend or revise the system in later stages.

Readers who want a general understanding of the system should begin with Section 2: Overall Description.

Readers who are interested in system integration and external dependencies should focus on Section 4: External Interface Requirements.

Readers who want to understand quality constraints and operational expectations should read Section 5: Other Nonfunctional Requirements.

1.4 Project Scope

The Mentor Caring System (MCS) is a web-based information system designed to support the **BNBU Mentor Caring Programme**, which aims to help undergraduate students adapt to university life more effectively through structured mentoring and follow-up activities. The system is intended to provide a centralized platform for managing mentoring-related information, communication, and records for the different participants involved in the programme.

The system supports multiple types of users, including **Administrator, Faculty Consultant**, Mentor, Student, MCP Coordinator, and optionally **Supporting Staff**. It is intended to improve the efficiency and consistency of mentoring management by digitalizing major mentoring-related activities.

The main goals of the system are to:

- allocate mentors;
- report and maintain interview records;
- support appointment arrangement between mentors and students;
- generate summary reports;
- track relevant user and mentoring information.

To achieve these goals, the system supports key mentoring management activities carried out by different user roles, including administrators, faculty consultants, mentors, students, and MCP coordinators. It is expected to improve the efficiency, consistency, and traceability of mentoring work by integrating mentor allocation, record management, appointment support, information search, and communication functions into one system. The system also helps preserve historical mentoring records and supports the import of organizational and mentoring data, which is important for maintaining continuity when mentoring assignments or group membership change.

From the universitys perspective, MCS supports the operational objectives of the BNBU Mentor Caring Programme by enabling more systematic management of mentoring activities, improving communication among participants, and making student support information easier to maintain and follow up. In this way, the system contributes to a more coordinated and scalable mentoring process within BNBU.

1.5 References

Design and Implementation of a Mentor Caring System, Project Long Description, Version 1.0, 2025–2026 Semester 2.

2. Overall Description

2.1 Product Perspective

The Mentor Caring System is a new information system developed for the BNBU Mentor Caring Programme. It is designed to support mentor caring activities for undergraduate students, including mentor allocation, record management, appointment arrangement, and communication.

The system is not intended to replace an existing product. It operates within the university environment and relies on existing institutional resources, especially the university account database for login and user identification, as well as related student and staff information sources.

The system also interacts with external supporting components, such as Excel files for data import and the university email or communication system for user notification. The context model in this section shows the system boundary and its main external interfaces.

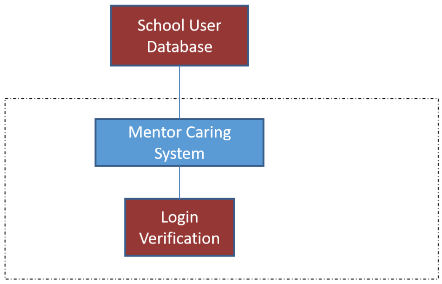

Figure 1: Context Model of the Mentor Caring System

2.2 Product Features

The Mentor Caring System supports the main activities of the BNBU Mentor Caring Programme. Its major features include role-based user access, mentor-group management, information search, interview record management, appointment arrangement, communication between users, special case forwarding, and record export. These functions help different users complete mentor caring tasks efficiently through one integrated system.

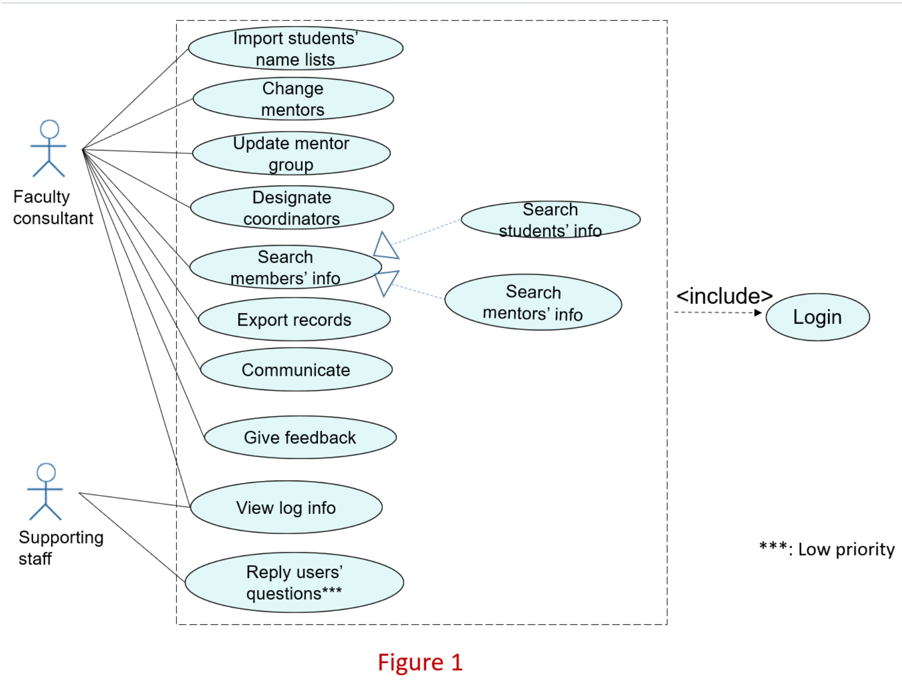

Figure 2: Use Case Diagram for Faculty Consultant and Supporting Staff

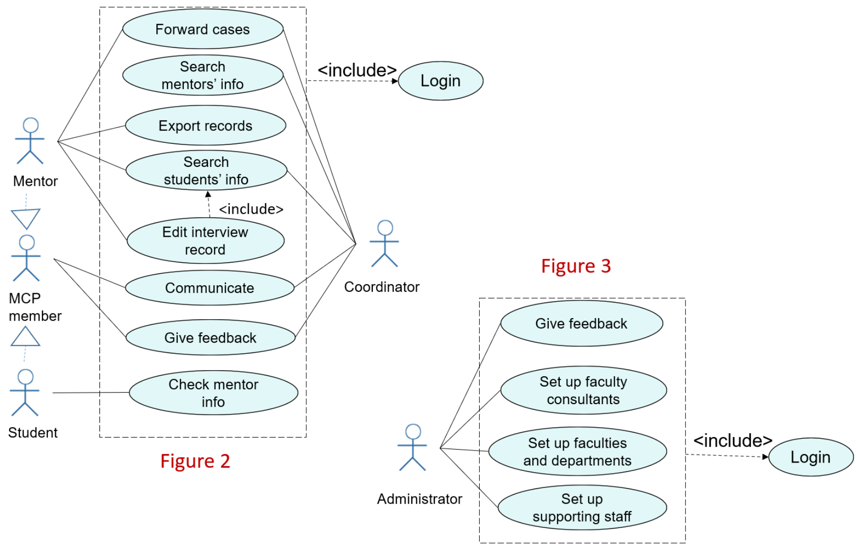

Figure 3: Use Case Diagram for Mentor, MCP Member, Student, Coordinator, and Administrator

2.2.1 Scenario for "Search Mentor Info"

Plan A – Basic Scenario

1. The user logs in (see the Login scenario).
2. The user activates the "Search Mentor Info" function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database for the group information.

6. The system displays the mentor's name associated with the group.
7. The system displays the record of one student in that group.

Plan B – Alternative Scenario: Group Not Found

1. The user logs in (see the Login scenario).
2. The user activates the "Search Mentor Info" function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database.
6. The system cannot find the group information.
7. The system displays a message indicating that the Group ID is invalid and the use case ends.

Plan C – Alternative Scenario: No Student Record

1. The user logs in (see the Login scenario).
2. The user activates the "Search Mentor Info" function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database for the group information.
6. The system finds the mentor information associated with the group.
7. The system cannot find any student record in that group.
8. The system displays the mentor's name and a message indicating that no student record is available.

2.3 User Classes and Characteristics

The main user classes of the Mentor Caring System are administrator, faculty consultant, mentor, student, MCP coordinator, and supporting staff.

- Administrator
  The administrator is a privileged user responsible for overall system setup and management. Usually there is only one administrator in the university. This user designates and manages faculty consultants and supporting staff. The administrator has the highest management authority but is expected to use the system less frequently than faculty consultants, mentors, and students.
- Faculty Consultant
  Faculty consultants are important management users at the faculty level. They are responsible for maintaining mentor allocation, managing mentoring groups, assigning MCP coordinators, searching information, and exporting records. They are frequent users, especially at the beginning of each academic year and whenever group allocation changes are needed.
- Mentor
  Mentors are core operational users of the system. They mainly use the system to view students in their own groups, maintain interview records, communicate with students, arrange interviews, and handle special cases. They are frequent users and

generally have moderate computer literacy. Their access rights are limited to their own groups.

- Student
  Students are end users of the system. Any current undergraduate student can log in to the system, but only students assigned to a mentoring group can continue to use its main functions. Students mainly use the system to check their mentors basic information and communicate with mentors. This user class is large in number and requires a simple and easy-to-use interface.
- MCP Coordinator
  MCP coordinators are department-level users designated by faculty consultants. They are responsible for searching mentor and student information, handling special cases, and communicating with related users. In some cases, an MCP coordinator may also serve as a mentor. This user class has moderate to high access privileges.
- Supporting Staff
  Supporting staff are optional users set up by the administrator. They mainly view student log information and may respond to user feedback about system usage. Since this function is optional, this user class is less important than the main operational user classes, and its usage frequency is expected to be relatively low.

2.4 Operating Environment

- The system depends on an existing school database that stores staff and student accounts, passwords, names, and roles.
- The system must support the import of Excel files for organization structure, mentor allocation, and MCP coordinator setup.
- The system must also support exporting information and interview records to Word files.
- The system must support the universitys email service to send email notification.

2.5 Design and Implementation Constraints

- The system shall use existing university accounts for user login rather than creating an independent account system.
- Import file format constraints: the system should strictly import file of Excel for setting faculties, departments, majors, student-mentor allocations, and MCP coordinators.
- Export file format constraints: the system should strictly export selected information and interview records to Word files.
- The system shall preserve historical student information even when mentors are changed or students are removed from groups.
- The system design shall be compatible with the existing school database and related university data sources.
- The system shall enforce role-based access restrictions for people with specific role could only access to specific functions and data scopes.

2.6 User Documentation

The following documentation will be delivered with the MCS:

- User Manual for general system use
- Administrator Guide for system administration
- Online Help integrated into the system
- Excel Template Files for required data import

2.7 Assumptions and Dependencies

- It is assumed that a university database containing valid staff and student accounts, passwords, names, and user roles is already available for authentication and identity management.
- It is assumed that faculty consultants will provide correct and properly formatted Excel files for setting up.
- It is assumed that student email addresses can be derived or obtained from student IDs.

3. System Features

3.1 Login

3.1.1 Description and Priority

Actor:

Faculty Consultant, Supporting Staff, Mentor, MCP Coordinator, Student, Administrator

Description:

This feature allows a user to log into the Mentor Caring System using a valid university account and password. After successful authentication, the system directs the user to the corresponding home page according to the users role. If a student has not yet been allocated to any mentoring group, the system displays a warning message and requires the student to quit the system.

Priority: High

3.1.2 Stimulus/Response Sequences

Assumption:

The user has a valid university account and password and is on the login page.

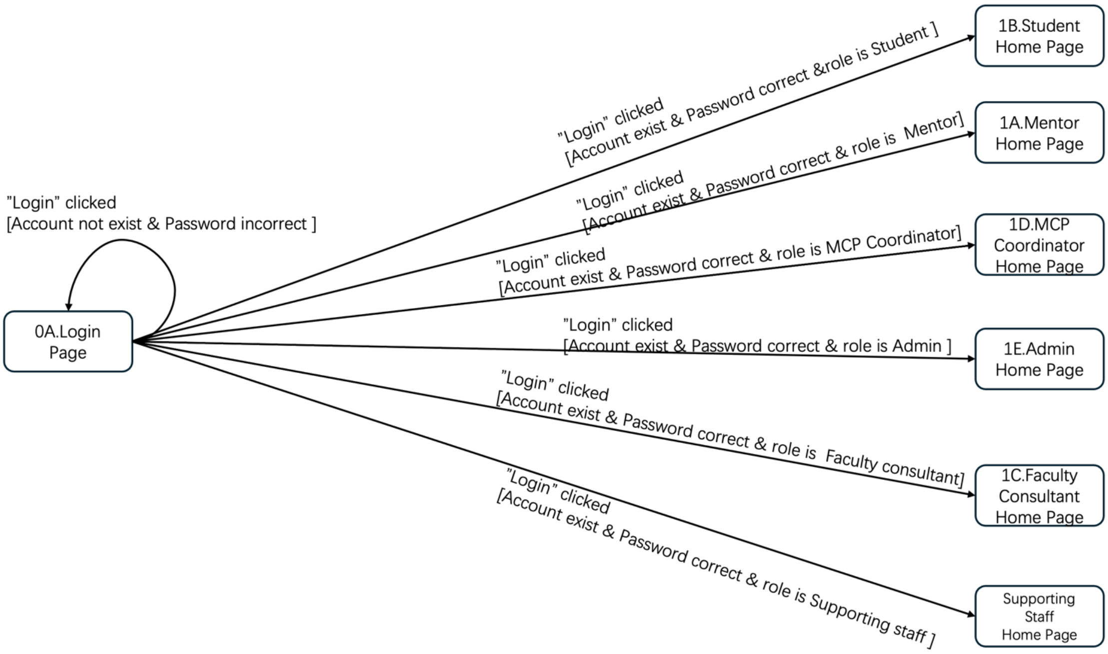

Figure 4: User login

Basic Scenario:

1. The user opens the Mentor Caring System login page.
2. The system displays the login page with fields for account and password.
3. The user enters the account and password.
4. The system accepts the entered login information.
5. The user clicks the Login button.
6. The system validates the account and password.
7. The system identifies the users role.
8. The system displays the corresponding home page.
9. The use case ends.

Alternative Scenario 1: Invalid account or password

1. The user opens the login page.
2. The system displays the login form.
3. The user enters an invalid account or password.
4. The system accepts the entered information.
5. The user clicks the Login button.
6. The system detects that the login information is invalid.
7. The system displays an error message and remains on the login page.

3.2 Search Students Info

3.2.1 Description and Priority

Actor:

Mentor, Coordinator, Faculty Consultant

Description:

This feature allows a Mentor or MCP Coordinator to search for a students information by entering the student ID. If the student record is found and the user is authorized to access it, the system shall display the students basic information and interview records. This feature serves as a prerequisite for Section 3.3 Edit Interview Record.

Priority: High

3.2.2 Stimulus/Response Sequences

Assumption:

The user has logged in and is authorized to use the search function.

Basic Scenario

1. The user selects Search Students Info from the home page.
2. The system displays the Search Students Info page.
3. The user enters a student ID and clicks Search.
4. The system validates the input and searches for the student record.
5. The user is authorized to access the student record.
6. The system displays the students basic information and interview records in read-only form.
7. The use case ends.

Alternative Scenario 1: Invalid or Empty Student ID

1. The user opens the Search Students Info page.
2. The system displays the student ID input field.
3. The user leaves the field empty or enters an invalid student ID and clicks Search.
4. The system displays a warning message.
5. The user remains on the Search Students Info page.
6. The use case ends.

Please see 3.3.2, Transition diagram is included in 3.3.2

3.2.3 Functional Requirements

REQ-1:

The system shall allow a Mentor or MCP Coordinator to search for student information by entering a student ID.

REQ-2:

The system shall validate the entered <u>student</u> ID before processing the search request.

REQ-3:

If the search is successful, the system shall display the students information, including student ID, group ID, mentor name, faculty/department/major, interview records, and status.

REQ-4:

The system shall allow a Mentor to access only the information of students in the mentors own mentoring groups.

REQ-5:

The system shall allow an MCP Coordinator to access only the information of students within the coordinators authorized department or scope.

REQ-6:

If no matching student record is found, the system shall display a clear error message.

REQ-7:

If the user is not authorized to access the requested student record, the system shall deny access and display a warning message.

REQ-8:

The system shall display the retrieved student information in read-only mode for this feature.

3.3 Edit Interview Record

3.3.1 Description and Priority

Actor:

Mentor

Description:

This feature allows a mentor to edit a students interview record after the students information has been found through the use case Search Students Info (More information of Search Students Info, check 3.2). A mentor must search for students information first. The mentor could click Edit to add, modify, and delete interview. The mentor could modify and delete an existing record, and mentor could enter a new record if the record is empty. For any changes a mentor made, Save and Cancel button is provided for mentor to decide.

Priority: High

3.3.2 Stimulus/Response Sequences

The combined state transition diagram for Search Students Info (3.2) and Edit Interview Record (3.3) is shown in Figure 5.

Assumption:

The user has already logged into the Mentor Caring System with a valid account.

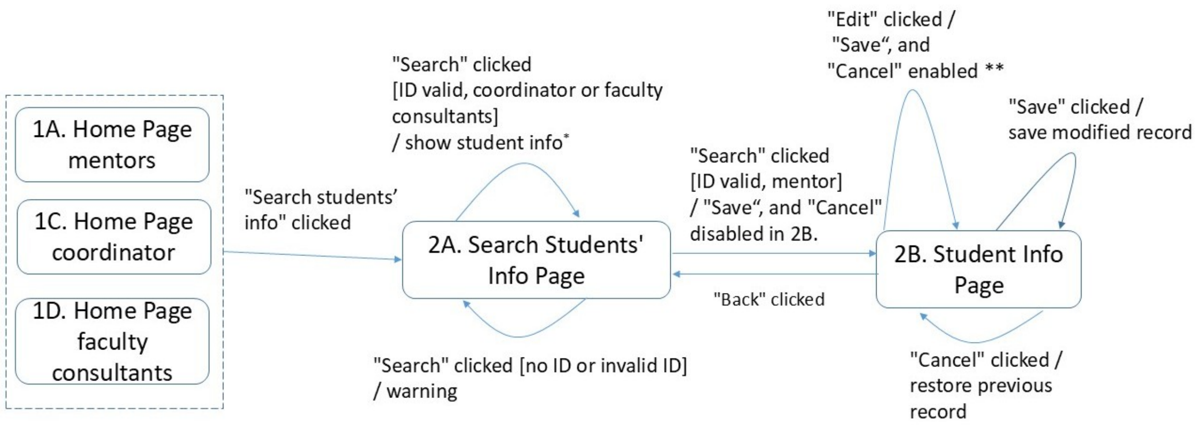

Figure 5: Search Student Information

Basic Scenario

1. The mentor views the Students' Info page.
2. The system displays the student's basic information and current interview records.
3. The mentor clicks Edit.
4. The system enables editing controls for interview records.
5. The mentor adds a new record or modifies or deletes an existing record.
6. The system keeps the changes in editable form and displays Save and Cancel controls.
7. The mentor clicks Save.
8. The system stores the updated interview record and refreshes the Student Info page with the latest content.
9. The use case ends.

Alternative Scenario 1: Cancel Editing

1. The mentor is editing an interview record.
2. The system displays the editable record and the Save and Cancel controls.
3. The mentor clicks Cancel.
4. The system discards all unsaved changes and restores the previous content.
5. The system returns to the Student Info page.
6. The use case ends.

Alternative Scenario 2: Unauthorized Role

1. A non-mentor user opens a student record.
2. The system displays the record in read-only mode.
3. The user attempts to edit the interview record.
4. The system denies the editing action and keeps the page in read-only mode.
5. The use case ends.

3.3.3 Functional Requirements

REQ-1:

The system shall allow only a mentor to edit interview records through this feature.

REQ-2:

The system shall require the students information to be displayed before the mentor can edit the students interview record.

REQ-3:

The system shall allow the mentor to add, modify, and delete an interview record for a student in the mentors own group.

REQ-4:

Each interview record shall contain the following items: interview date and time, problem statement, interview summary, and follow-up actions.

3.4 Search Mentor Info

3.4.1 Description and Priority

Actor:

Faculty Consultant, Coordinator

Description:

This feature allows a user to search for mentor information by entering a Group ID. The system displays the mentors name associated with the group and shows student records in that group.

Priority: High

3.4.2 Stimulus/Response Sequences

Assumption:

The Faculty Consultant or MCP Coordinator has already logged into the system and is authorized to access mentor information.

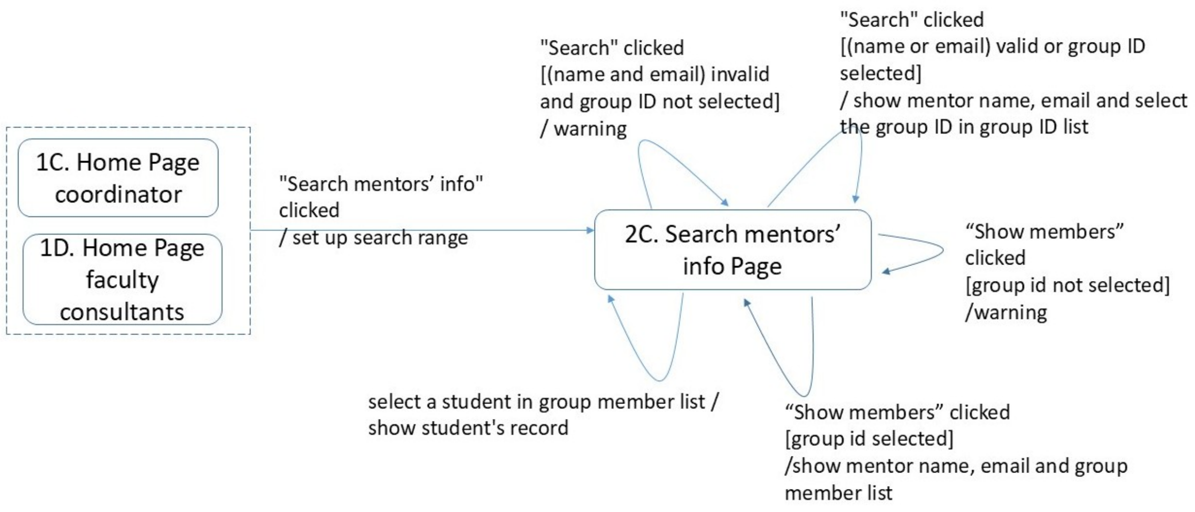

Figure 6: Search Mentor's Information

Basic Scenario

1. The user selects Search Mentor Info from the home page.
2. The system displays the Mentor Search page.
3. The user enters a mentor name or email, or group ID and clicks Search.
4. The system displays a list of matching mentors, including their names, emails, and associated group IDs.
5. The user selects a group ID and clicks Show Members.
6. The system displays the mentor information together with the member list of the selected group.
7. The user selects a student name from the member list.
8. The system displays the selected students detailed record.
9. The use case ends.

Alternative Scenario 1: Invalid Search Input or No Result Found

1. The user opens the Mentor Search page.
2. The system displays the search fields.
3. The user enters invalid search information or information that does not match any record and clicks Search.
4. The system displays an error message.
5. The user remains on the Mentor Search page.
6. The use case ends.

Alternative Scenario 2: No Group Selected

1. The user searches for mentor information successfully.
2. The system displays the mentor information and the available group ID list.
3. The user clicks Show Members without selecting a group ID.
4. The system displays a warning message.
5. The user remains on the Mentor Result page.
6. The use case ends.

Alternative Scenario 3: Navigate Back and Search Again

1. The user is on the Mentor Result Page, Member List Page, or Student Record Page.
2. The system displays the current page and navigation controls.
3. The user clicks Search Again or Back.
4. The system returns to the previous page according to the navigation path.
5. The use case continues or ends.

3.4.3 Functional Requirements

REQ-1: The system shall allow users to search for mentor information using name, email, or group ID.

REQ-2: The system shall display mentor name and group ID in the search results.

REQ-3: The system shall allow users to view group members by selecting a group ID.

REQ-4: The system shall allow users to view student records by selecting a student name.

3.5 Import students name lists

3.5.1 Description and Priority

Actor:

Faculty Consultant

Description:

This feature allows a faculty consultant to import a student-mentor allocation Excel file into the Mentor Caring System. The imported file contains student and mentor allocation data, such as student ID, student name, major, status, group ID, mentor name, and mentor email. After a valid file is uploaded, the system shall process the file and update the corresponding student-mentor allocation information. Existing historical student information shall be preserved during the yearly allocation import process.

Priority: Medium

Assumption:

The user has already logged into the Mentor Caring System with a valid account and has the authority to perform Excel import operations. The uploaded file follows the required Excel template format.

3.5.2 Stimulus/Response Sequences

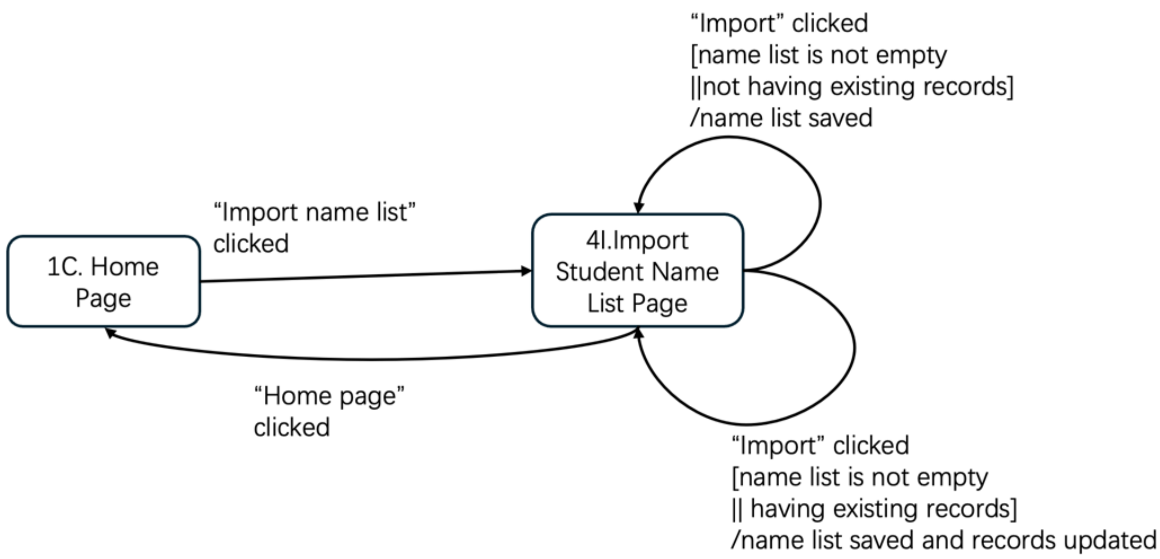

Figure 7: Import Students' Name List

Basic Scenario:

1. The faculty consultant selects the Import students name lists function.
2. The system displays the import page and provides the option to upload an Excel file.
3. The faculty consultant chooses a valid Excel file for student-mentor allocation import.

4. The system accepts the uploaded file and checks its format.
5. The faculty consultant confirms the import request.
6. The system validates the file content and reads the allocation data.
7. The system updates the student-mentor allocation information in the database.
8. The system preserves existing historical student information during the import process.
9. The system displays a success message indicating that the import is completed.
10. The use case ends.

Alternative Scenario 1: Invalid File Format

1. The faculty consultant selects the import function.
2. The system displays the file upload page.
3. The faculty consultant chooses an invalid file or a file in the wrong format.
4. The system checks the file and detects that the format is invalid.
5. The system displays an error message and asks the user to upload a valid Excel file.

Alternative Scenario 2: Invalid or Incomplete Data in File

1. The faculty consultant selects the import function.
2. The system displays the file upload page.
3. The faculty consultant uploads an Excel file.
4. The system validates the file content.
5. The system detects invalid, incomplete, or inconsistent data in the file.
6. The system displays an error message indicating the problem.
7. The system cancels the import operation or asks the user to correct the file before re-importing.

Alternative Scenario 3: Duplicate Data in File

1. The faculty consultant selects the import function.
2. The system displays the file upload page.
3. The faculty consultant uploads a valid Excel file.
4. The system validates the file content and detects duplicated data.
5. The system keeps the existing valid information unchanged.
6. The system displays a warning message indicating that duplicate data have been found and handled.
7. The use case ends.

3.6 Designate coordinators

3.6.1 Description and Priority

Actor:

Faculty Consultant

Description:

This feature allows a faculty consultant to import an Excel file for setting MCP coordinators in the Mentor Caring System. The imported file contains the coordinator name, email, and department. After a valid file is uploaded, the system shall process the file and assign the corresponding MCP coordinator to each department.

Priority: Medium

3.6.2 Stimulus/Response Sequences

Assumption:

The user has already logged into the Mentor Caring System with a valid account and has the authority to import MCP coordinator setup data. The uploaded file follows the required Excel template format.

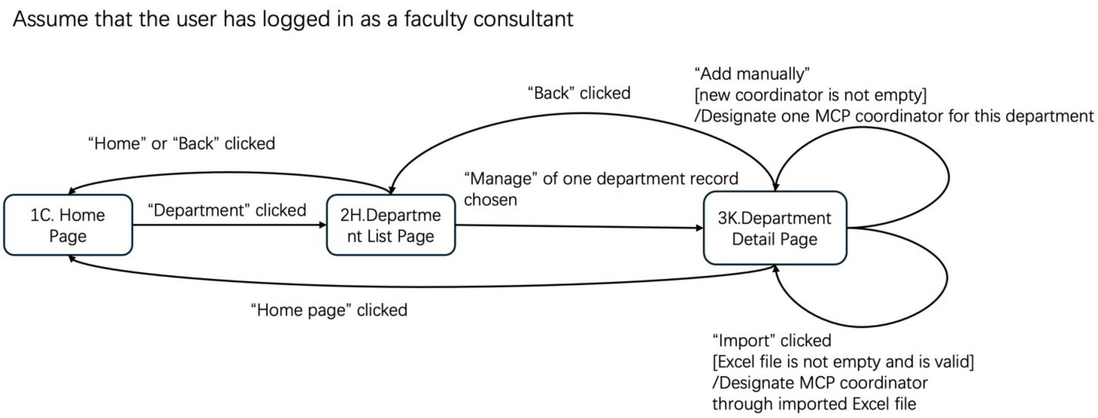

Figure 8: Designate Coordinators

Basic Scenario

1. The faculty consultant clicks the Department option from the Home Page.
2. The system displays the Department List Page.
3. The faculty consultant selects a department record.
4. The system displays the Department Detail Page.
5. The faculty consultant chooses to designate an MCP coordinator.
6. The system allows the user to either manually assign a coordinator or import coordinator information through an Excel file.
7. The faculty consultant provides the coordinator information.
8. The system accepts the input or uploaded file and validates the data.
9. The faculty consultant confirms the designation.
10. The system assigns the MCP coordinator to the selected department and updates the database.
11. The system displays a success message.
12. The use case ends.

Alternative Scenario 1: Invalid or Incomplete Data

1. The faculty consultant reaches the Department Detail Page.
2. The faculty consultant enters invalid or incomplete coordinator information or uploads an incorrect file.
3. The system validates the input and detects an error.
4. The system displays an error message and requests correction.

Alternative Scenario 2: Unauthorized Access

1. A user selects the Department function.
2. The system checks the user role.
3. The system determines that the user is not authorized.
4. The system denies access and displays a warning message.

Alternative Scenario 3: No Department Selected

1. The faculty consultant opens the Department List Page.
2. The system displays the list of departments.
3. The faculty consultant attempts to proceed without selecting a department.
4. The system displays a warning message asking the user to select a department.

3.7 Set up faculties and departments

3.7.1 Description and Priority

Actor:

Administrator

Description:

This feature allows the administrator to set up the faculty, department, and major information in the Mentor Caring System. The administrator may enter the organization information manually or import it through an Excel file. The system shall validate the input and update the organization information accordingly.

Priority: high

3.7.2 Stimulus/Response Sequences

Assumption:

The administrator has already logged into the Mentor Caring System with a valid account and has the authority to manage faculty, department, and major information.

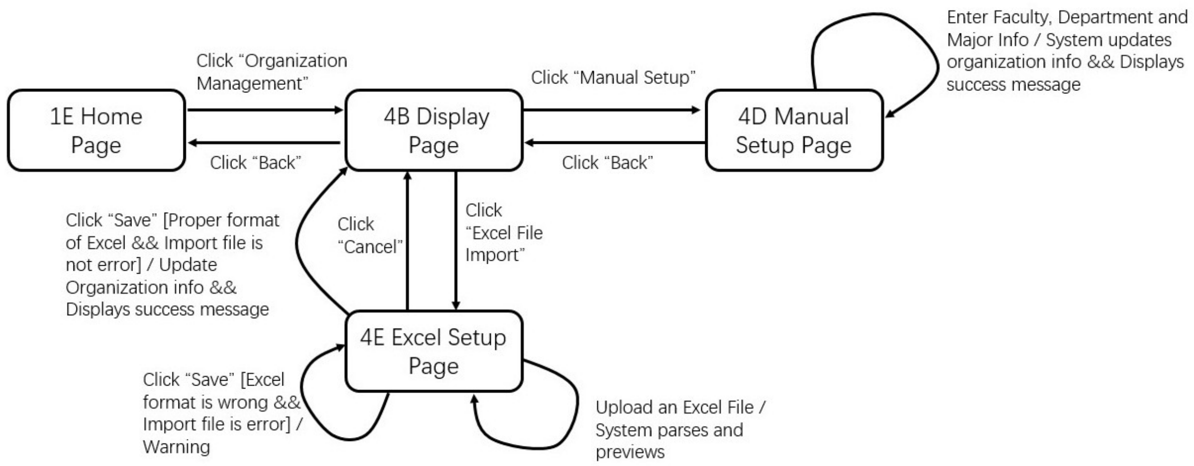

Figure 9: set Up Faculty and Department

Basic Scenario:

1. The administrator clicks the Organization Management function from the Home Page.
2. The system displays the Display Page.
3. The administrator chooses to set up the organization information manually or through Excel file import.
4. The system displays the corresponding Manual Setup Page or Excel Setup Page.
5. The administrator enters the faculty, department, and major information manually, or uploads an Excel file.
6. The system accepts the input or uploaded file.
7. The administrator clicks the Save button.
8. The system validates the input information or file content.
9. The system updates the organization information.
10. The system displays a success message.
11. The use case ends.

Alternative Scenario 1: Invalid File Format

1. The administrator selects the setup function.
2. The system displays the setup page.
3. The administrator chooses to upload a file.
4. The system provides the file upload option.
5. The administrator uploads a file in an invalid format.
6. The system detects that the file format is invalid.
7. The system displays an error message and asks the administrator to upload a valid file.

Alternative Scenario 2: Invalid or Incomplete Data

1. The administrator selects the setup function.
2. The system displays the setup page.
3. The administrator enters the information manually or uploads a file.
4. The system validates the content.
5. The system detects invalid or incomplete faculty or department information.
6. The system displays an error message and asks the administrator to correct the data.
7. The setup operation is not completed.

Alternative Scenario 3: Unauthorized Access

1. A user selects the Set up faculties and departments function.
2. The system checks the user role.
3. The system finds that the user is not authorized to perform this operation.
4. The system denies access and displays a warning message.
5. The use case ends.

3.8 Export records

3.8.1 Description and Priority

Actor:

Mentor

Description:

This feature allows a mentor to export interview records in the Mentor Caring System. The mentor may choose to export all student interview records or only the selected student records in the mentoring group. After the export request is confirmed, the system shall generate a Word file, download it to the client side, and display the export result.

Priority: High

3.8.2 Stimulus/Response Sequences

Assumption:

The mentor has already logged into the Mentor Caring System with a valid account and has the authority to access the records of students in the mentoring group.

Assume that the user has logged in as a faculty consultant

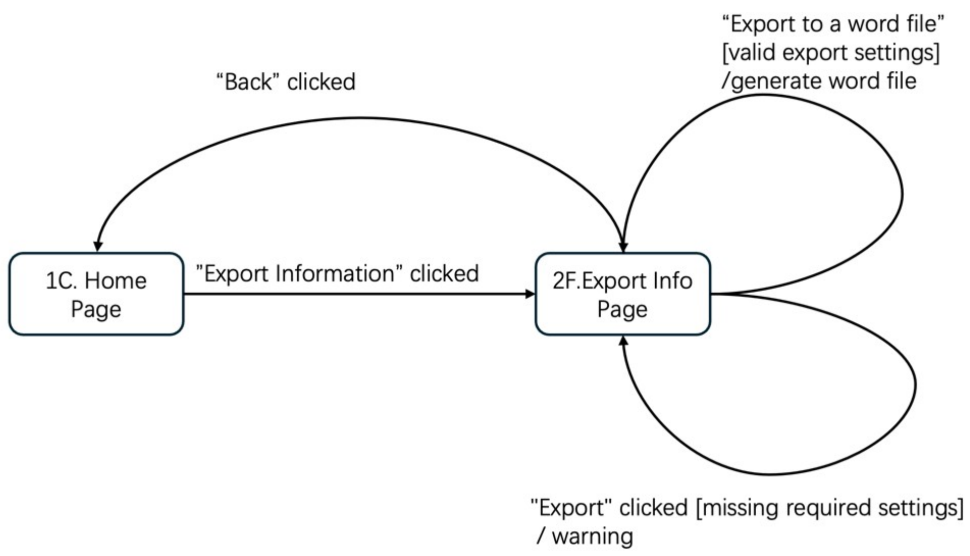

Figure 10: Export Record of Faculty

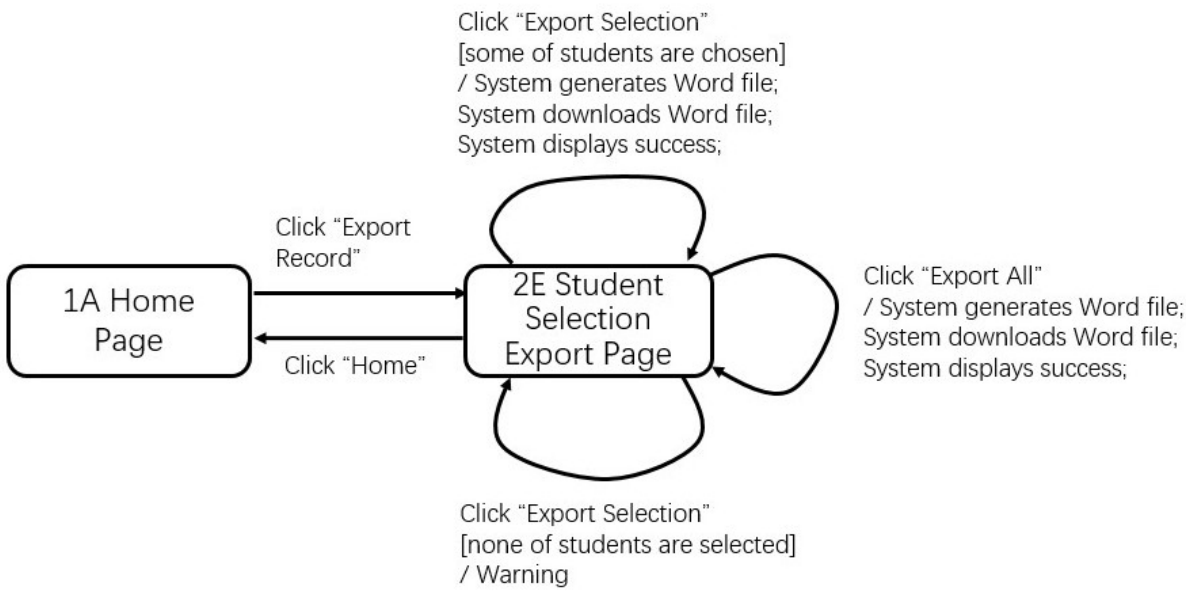

Figure 11: Export Record of Mentor

Basic Scenario:

1. The mentor clicks the Export Record function from the Home Page.
2. The system displays the Student Selection Export Page.
3. The mentor chooses to export all records or selected student records.
4. The system accepts the export choice.
5. The mentor clicks Export All or Export Selection.
6. The system generates the Word file.
7. The system downloads the Word file.
8. The system displays a success message.
9. The use case ends.

Alternative Scenario 1: No record selected

1. The mentor selects the Export records function.
2. The system displays the export page.
3. The mentor does not select any record and submits the request.
4. The system detects that no record has been selected.
5. The system displays a warning message and asks the mentor to select records before exporting.

Alternative Scenario 2: No matching records found

1. The mentor selects the Export records function.
2. The system displays the export page.
3. The mentor selects the target records and confirms the request.
4. The system searches for the corresponding records.
5. The system finds that no matching records are available.
6. The system displays a message indicating that no records can be exported.

Alternative Scenario 3: Export generation failure

1. The mentor selects the Export records function.

2. The system displays the export page.
3. The mentor selects the records and confirms the request.
4. The system starts to generate the export file.
5. The system fails to complete the export process.
6. The system displays an error message and asks the mentor to try again later.

3.9 Check mentor info

3.9.1 Description and Priority

Actor:

Student

Description:

This feature allows a student to check the information of the assigned mentor in the Mentor Caring System. The student can view the mentors basic information, such as name, office, email, and other relevant contact details. The system shall display the mentor information associated with the students mentoring group.

Priority: high

3.9.2 Stimulus/Response Sequences

Assumption:

The student has already logged into the Mentor Caring System with a valid account and has been assigned to a mentoring group.

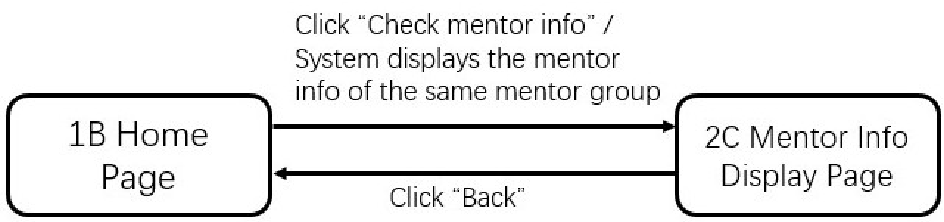

Figure 12: Check Mentor’s Information

Basic Scenario:

1. The student clicks the Check mentor info function from the Home Page.
2. The system displays the Mentor Info Display Page.
3. The system shows the mentor information of the same mentor group.
4. The student views the mentor information.
5. The use case ends.

Alternative Scenario 1: Return to Home Page

1. The student opens the Mentor Info Display Page.
2. The student clicks the Back button.

3. The system returns to the Home Page.

Alternative Scenario 2: No Mentor Information Available

1. The student clicks the Check mentor info function from the Home Page.
2. The system opens the Mentor Info Display Page.
3. The system cannot find the mentor information of the same mentor group.
4. The system displays a warning message.
5. The system remains on the Mentor Info Display Page.

3.10 Communicate

3.10.1 Description and Priority

Actor:

Faculty Consultant, Mentor, Student, Coordinator

Description:

This feature allows users to communicate with other authorized users in the Mentor Caring System. A user may select a receiver, enter the message content, and send the message through the system. The receiver can log into the system and read the received message.

Priority:

High

3.10.2 Stimulus/Response Sequences

Assumption:

The user has logged into the system as a faculty consultant, mentor, MCP coordinator, or student and has permission to communicate with other users.

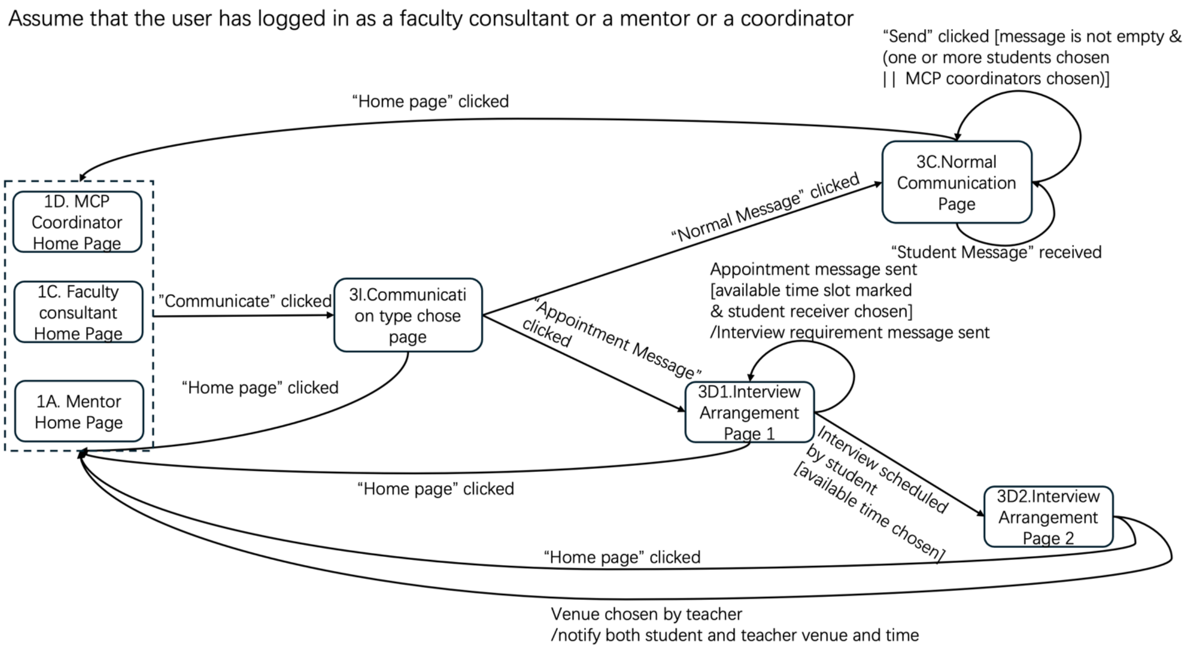

Figure 13: Communicate

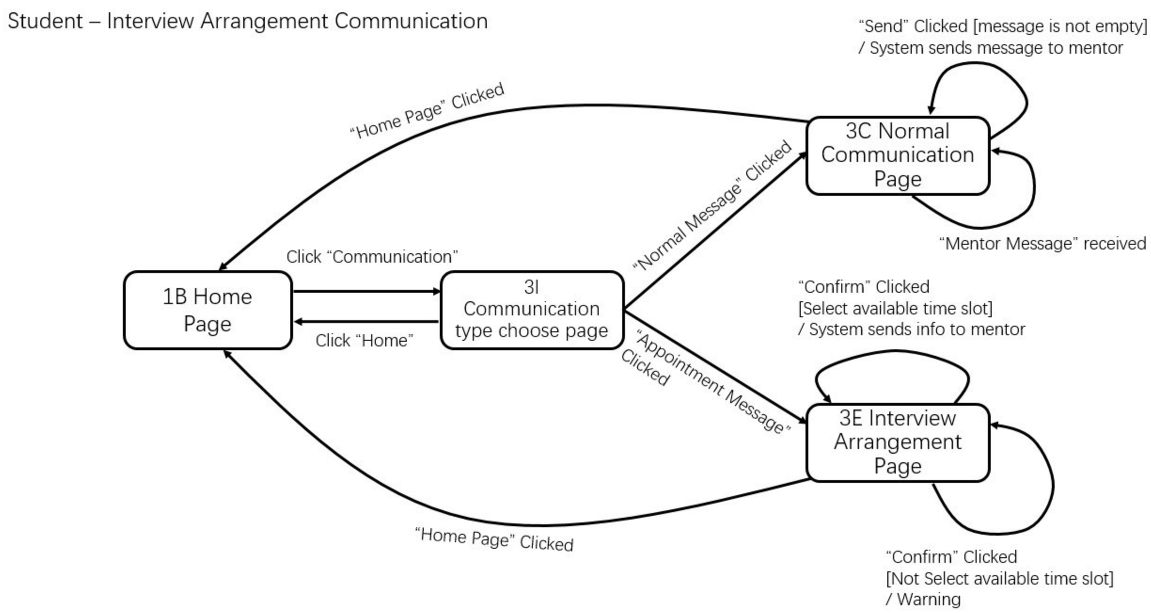

Figure 14: Student Interview Arrangement Communication

Basic Scenario: Normal Message Communication

1. The user clicks the "Communicate" function from the Home Page.
2. The system displays the Communication Type Selection Page.
3. The user selects the Normal Message option.
4. The system displays the Normal Communication Page.

5. The user selects one or more receivers.
6. The system displays the message input area.
7. The user enters the message content.
8. The system accepts the entered message content.
9. The user clicks the Send button.
10. The system validates that the message is not empty and that valid receivers are selected.
11. The system sends the message and stores it in the database.
12. The system marks the message as unread for the receiver.
13. The receiver logs into the system and views the message.
14. The system displays the received message.
15. The use case ends.

Basic Scenario 2: Interview Arrangement (Appointment Communication)

Step A: Initiate an appointmentMentor / Coordinator

1. The user selects the Communicate function.
2. The system displays the Communication Type Selection Page.
3. The user selects the "Appointment Message" option.
4. The system displays the Interview Arrangement Page (Page 1).
5. The user selects an available time slot and a student receiver.
6. The system accepts the selected time slot and receiver.
7. The system sends an appointment message to the student.
8. The system sends an interview requirement message together with the appointment request.

Step B: Student confirmation timeStudent side

1. The student receives the appointment message.
2. The student logs into the system and opens the Interview Arrangement Page.
3. The student selects an available time slot.
4. The student clicks the Confirm button.
5. The system sends the confirmed schedule to the mentor.

Step C: The supervisor confirms the location.Mentor side

1. The mentor receives the confirmed schedule.
2. The system displays the Interview Arrangement Page (Page 2).
3. The mentor selects the interview venue.
4. The system stores the interview time and venue.
5. The system notifies both the student and the mentor of the final interview arrangement.
6. The use case ends.

Alternative Scenario 1: Empty Message Content

1. The user selects the "Normal Message" option.
2. The system displays the message input area.
3. The user clicks "Send" without entering content.
4. The system detects that the message is empty.
5. The system displays a warning message.

Alternative Scenario 2: No Receiver Selected

1. The user enters message content.
2. The user clicks Send without selecting any receiver.
3. The system detects that no receiver is selected.
4. The system displays a warning message.

Alternative Scenario 3: No Time Slot Selected

1. The student opens the Interview Arrangement Page.
2. The system displays available time slots.
3. The student clicks "Confirm" without selecting a time slot.
4. The system displays a warning message.

Alternative Scenario 4: Time Slot Conflict

1. The user selects a time slot.
2. The system checks availability.
3. The system detects a scheduling conflict.
4. The system displays a warning message and asks the user to choose another slot.

3.11 Forward cases

3.11.1 Description and Priority

Actor:

Mentor, MCP Member

Description:

This feature allows a mentor or an MCP member to forward a student case to the next responsible user in the Mentor Caring System for further handling. The user can select the relevant student case, enter the case description, and submit the case through the system. After the request is confirmed, the system shall forward the case and store the related information.

Priority:

High

3.11.2 Stimulus/Response Sequences

Assumption:

The user has already logged into the Mentor Caring System with a valid account and has the authority to forward the selected case.

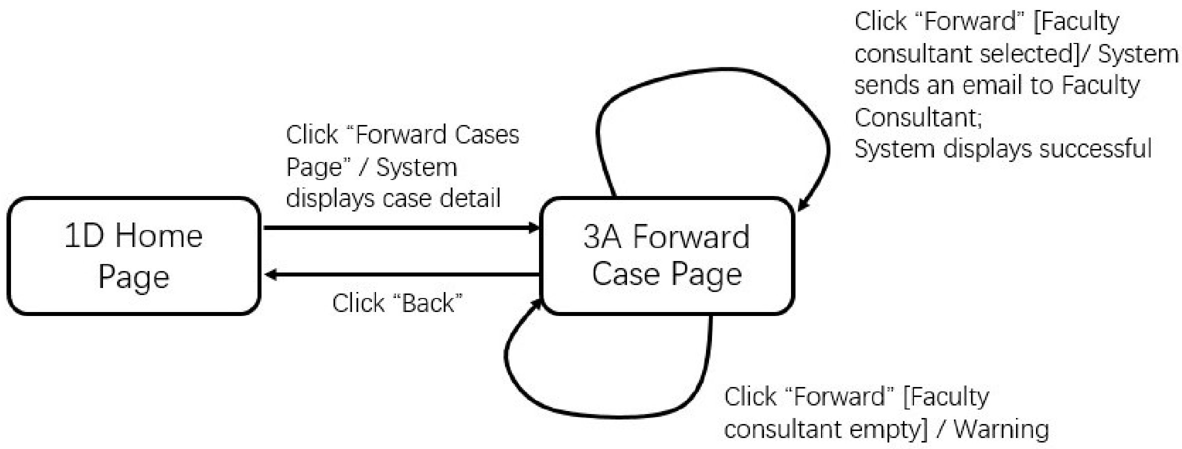

Figure 16: forward cases of coordinator

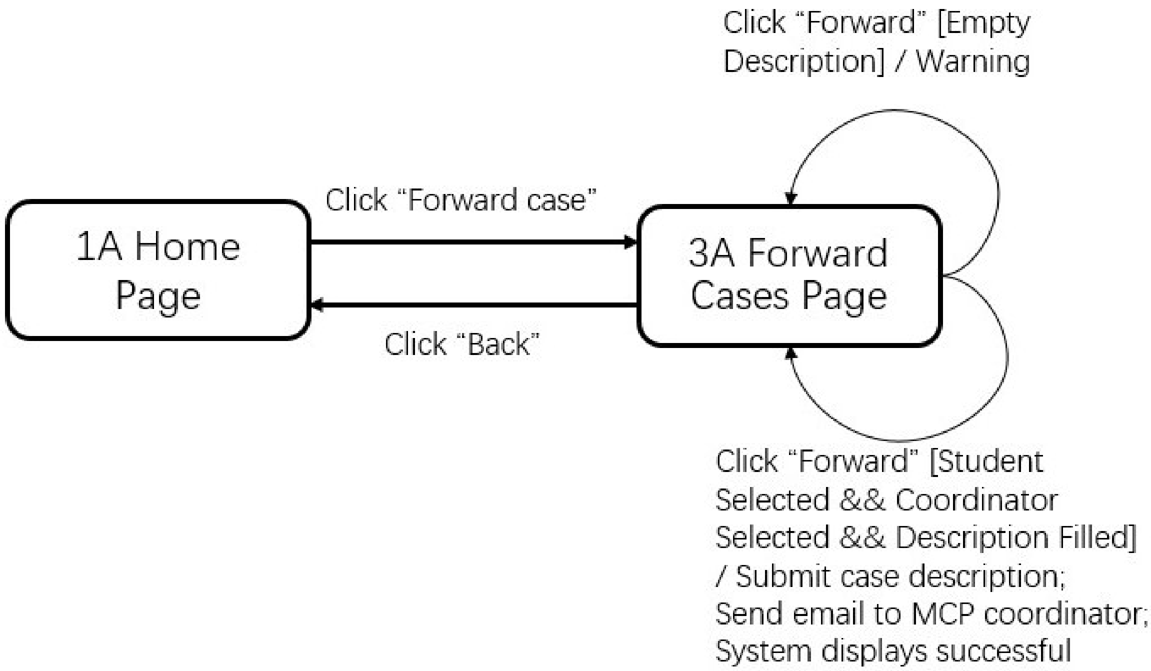

Figure 15: forward case of mentor

Basic Scenario:

1. The mentor clicks the Forward case function from the Home Page.
2. The system displays the Forward Cases Page.
3. The mentor selects a student case.
4. The system displays the case forwarding form.
5. The mentor selects a coordinator and enters the case description.
6. The system accepts the entered information.
7. The mentor clicks the Forward button.
8. The system submits the case description.

9. The system sends an email to the MCP coordinator.
10. The system displays a success message.
11. The use case ends.

Alternative Scenario 1: Empty Description

1. The mentor opens the Forward Cases Page.
2. The mentor selects a student case and a coordinator.
3. The mentor clicks the Forward button without entering the case description.
4. The system detects that the description is empty.
5. The system displays a warning message.
6. The system remains on the Forward Cases Page.

Alternative Scenario 2: Return to Home Page

1. The mentor opens the Forward Cases Page.
2. The mentor clicks the Back button.
3. The system returns to the Home Page.

Alternative Scenario 3: Coordinator Forwards Case to Faculty Consultant

1. The coordinator clicks the Forward Cases Page function from the Home Page.
2. The system displays the case detail on the Forward Case Page.
3. The coordinator selects a faculty consultant.
4. The system accepts the selected faculty consultant information.
5. The coordinator clicks the Forward button.
6. The system sends an email to the faculty consultant.
7. The system displays a success message.
8. The use case ends.

Alternative Scenario 4: No Faculty Consultant Selected

1. The coordinator opens the Forward Case Page.
2. The coordinator clicks the Forward button without selecting a faculty consultant.
3. The system detects that no faculty consultant has been selected.
4. The system displays a warning message.
5. The system remains on the Forward Case Page.

3.12 Change mentors and update mentor group

3.12.1 Description and Priority

Actor:

Faculty consultant

Description:

This feature allows a faculty consultant to modify mentor assignments and update group members.

Priority: medium

3.12.2 Stimulus/Response Sequences

Assumption:

The faculty consultant has logged in and has the authority to manage mentor groups.

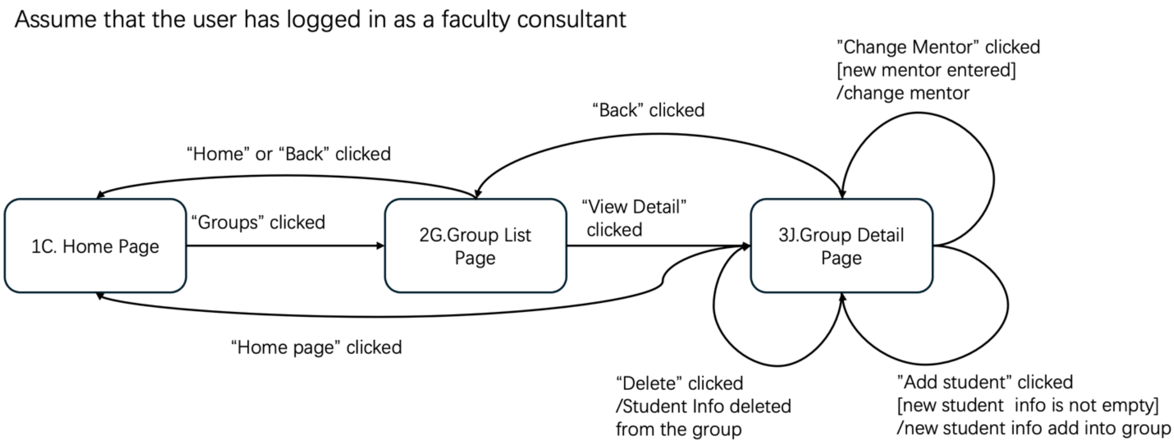

Figure 17: Change mentors and update mentor group

Basic Scenario:

1. The faculty consultant selects the Change mentors and update mentor group function.
2. The system displays the group list page.
3. The faculty consultant selects a group.
4. The system displays the group detail page.
5. The faculty consultant modifies the mentor assignment or updates group members.
6. The system accepts the changes.
7. The faculty consultant confirms the update.
8. The system updates the group information in the database.
9. The system displays a success message.
10. The use case ends.

Alternative Scenario 1: Invalid Update

1. The faculty consultant selects a group.
2. The system displays the group detail page.
3. The faculty consultant enters invalid or incomplete information.
4. The system detects invalid input.
5. The system displays an error message and requests correction.

Alternative Scenario 2: Unauthorized Access

1. A user selects the function.
2. The system checks the user role.
3. The system finds the user is not authorized.
4. The system denies access and displays a warning.

3.13 View log information

3.13.1 Description and Priority

Actor:

Faculty consultant, Supporting staff

Description:
This feature allows a faculty consultant and supporting staff to View log information.
Priority: medium

3.13.2 Stimulus/Response Sequences

Assumption:
The user has logged in and has permission to view log information.

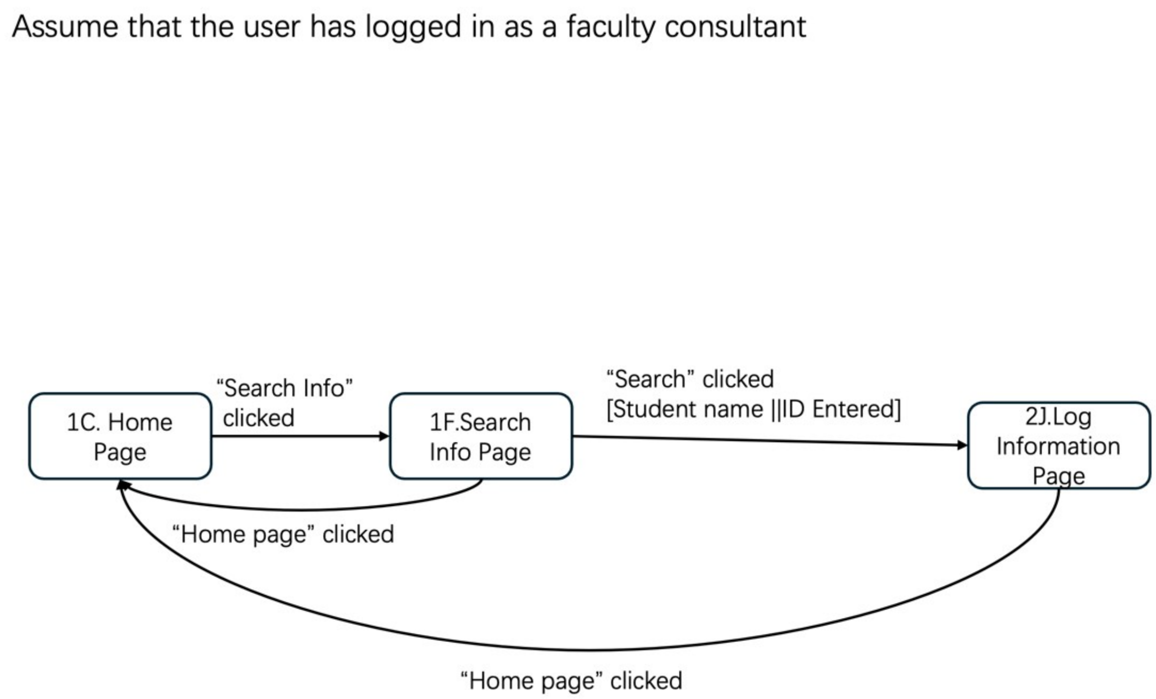

Figure 18: View log information

Basic Scenario:
1. The user selects the View log information function.
2. The system displays the log information page.
3. The user selects filter conditions (e.g., user, time range).
4. The system retrieves matching log records.
5. The system displays the log information.
6. The user reviews the log information.
7. The use case ends.

Alternative Scenario 1: No Log Found
1. The user selects filter conditions.
2. The system searches for matching logs.
3. The system finds no matching records.
4. The system displays a message indicating no log information is available.

Alternative Scenario 2: Unauthorized Access

1. A user selects the function.
2. The system checks the user role.
3. The system finds the user is not authorized.
4. The system denies access and displays a warning message.

3.14 Give feedback

3.14.1 Description and Priority

Actor:

Mentor, Student, MCP Coordinator, Faculty Consultant, Administrator

Description:

This feature allows authorized users to submit feedback through the Mentor Caring System. The user can enter feedback content on the Give Feedback Page and send it to the system. The system shall validate the feedback content and display the corresponding result message.

Priority: Medium

3.14.2 Stimulus/Response Sequences

Assumption: The user has already logged into the Mentor Caring System with a valid account and has the authority to access the feedback function.

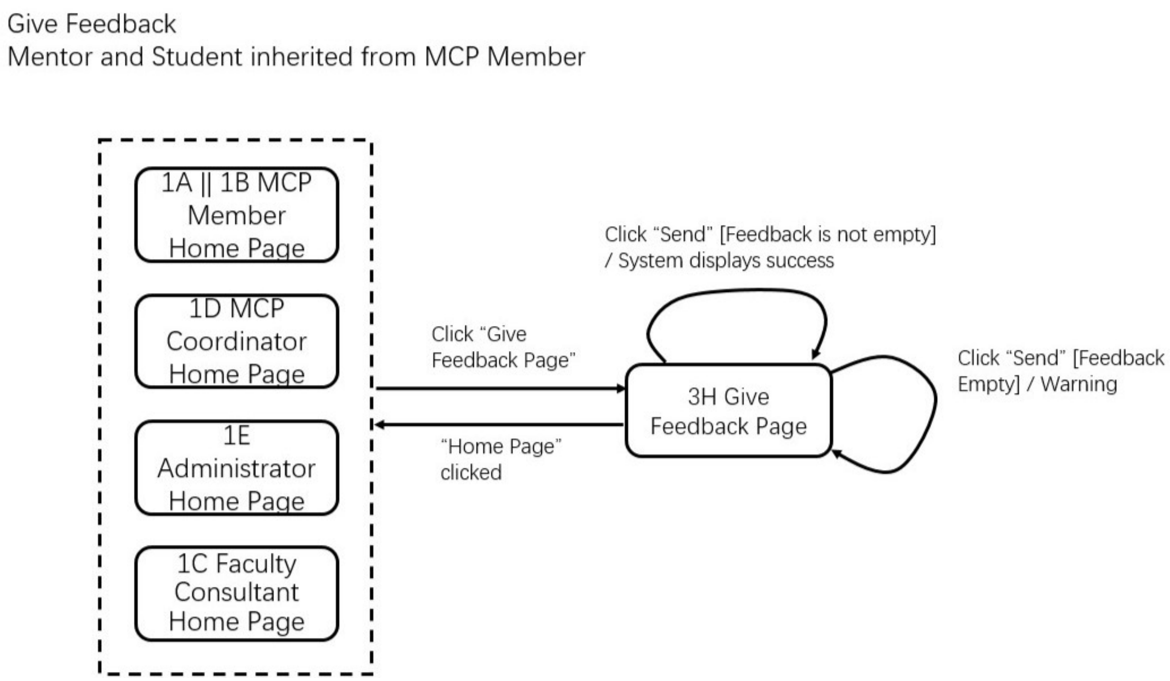

Figure 19: Give Feedback

Basic Scenario:

1. The user clicks the Give Feedback function from the Home Page.

2. The system displays the Give Feedback Page.
3. The user enters the feedback content.
4. The system accepts the entered feedback content.
5. The user clicks the Send button.
6. The system validates the feedback content.
7. The system submits the feedback successfully.
8. The system displays a success message.
9. The use case ends.

Alternative Scenario 1: Empty feedback

1. The user opens the Give Feedback Page.
2. The user clicks the Send button without entering any feedback content.
3. The system detects that the feedback is empty.
4. The system displays a warning message.
5. The system remains on the Give Feedback Page.

Alternative Scenario 2: Return to Home Page

1. The user opens the Give Feedback Page.
2. The user clicks the Home Page button.
3. The system returns to the corresponding Home Page.

3.15 Set up faculty consultants

3.15.1 Description and Priority

Actor:

Administrator

Description: This feature allows the administrator to manage faculty consultant information in the Mentor Caring System. The administrator can view the consultant list, add a new consultant, change existing consultant information, or delete a selected consultant. The system shall validate the input and update the consultant information accordingly.

Priority: high

3.15.2 Stimulus/Response Sequences

Assumption: The administrator has already logged into the Mentor Caring System with a valid account and has the authority to manage faculty consultant information.

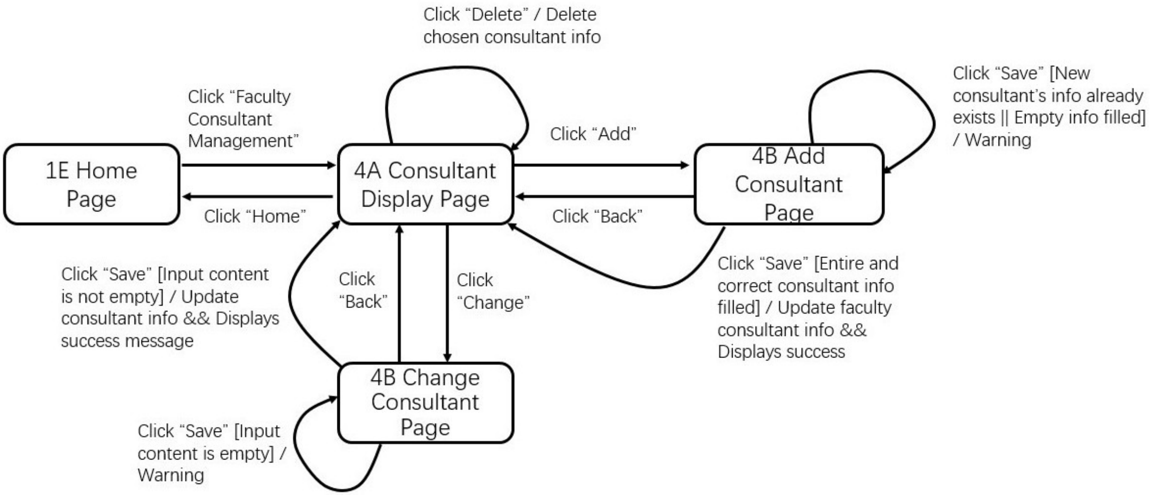

Figure 20: Set up faculty consultants

Basic Scenario:

1. The administrator clicks the Faculty Consultant Management function from the Home Page.
2. The system displays the Consultant Display Page.
3. The administrator chooses to add a new consultant or change an existing consultant.
4. The system displays the corresponding Add Consultant Page or Change Consultant Page.
5. The administrator enters or updates the consultant information.
6. The system accepts the input.
7. The administrator clicks the Save button.
8. The system validates the input information.
9. The system updates the faculty consultant information.
10. The system displays a success message.
11. The use case ends.

Alternative Scenario 1: Empty input when adding a consultant

1. The administrator opens the Add Consultant Page.
2. The administrator clicks the Save button without filling in the required information.
3. The system detects that the input content is empty.
4. The system displays a warning message.
5. The system remains on the Add Consultant Page.

Alternative Scenario 2: Invalid or duplicate consultant information

1. The administrator opens the Add Consultant Page.
2. The administrator enters incomplete information or information that already exists.
3. The administrator clicks the Save button.
4. The system detects that the consultant information is invalid, incomplete, or duplicated.
5. The system displays a warning message.
6. The system remains on the Add Consultant Page.

Alternative Scenario 3: Empty input when changing consultant information

1. The administrator opens the Change Consultant Page.
2. The administrator clicks the Save button without entering valid updated information.
3. The system detects that the input content is empty.
4. The system displays a warning message.
5. The system remains on the Change Consultant Page

Alternative Scenario 4: Delete consultant information

1. The administrator opens the Consultant Display Page.
2. The administrator selects a consultant record.
3. The administrator clicks the Delete button.
4. The system deletes the chosen consultant information.
5. The system updates the consultant list.
6. The use case ends.

Alternative Scenario 5: Return without saving

1. The administrator opens the Add Consultant Page or Change Consultant Page.
2. The administrator clicks the Back button.
3. The system returns to the Consultant Display Page.

3.16 Set up supporting staff

3.16.1 Description and Priority

Actor:

Administrator

Description: This feature allows the administrator to manage supporting staff information in the Mentor Caring System. The administrator can view the supporting staff list, add new supporting staff information, modify existing supporting staff information, or delete selected supporting staff records. The system shall validate the input and update the supporting staff information accordingly.

Priority:

high

3.16.2 Stimulus/Response Sequences

Assumption: The administrator has already logged into the Mentor Caring System with a valid account and has the authority to manage supporting staff information.

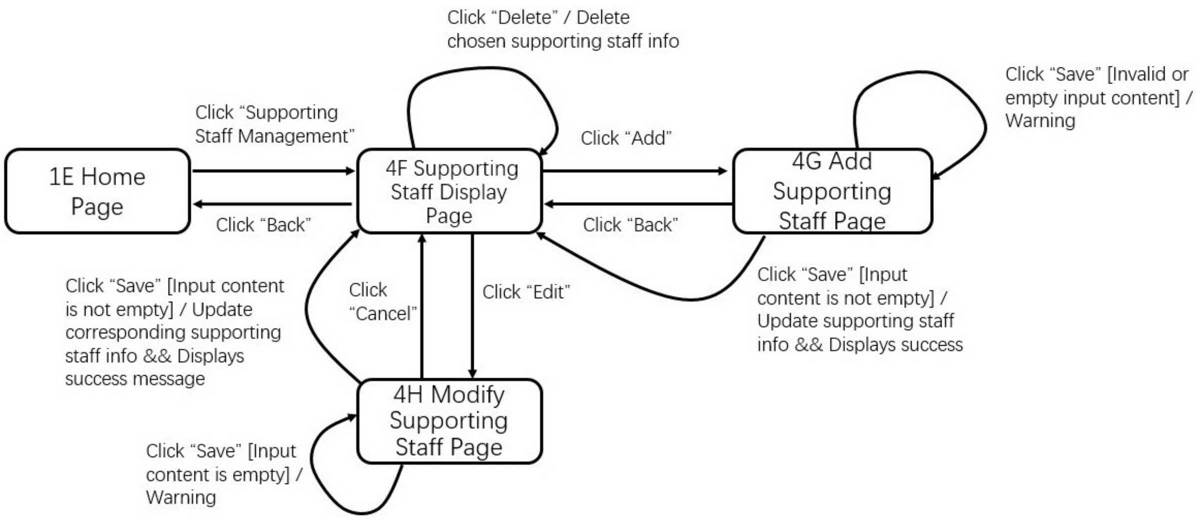

Figure 21: Set Up Supporting Staff

Basic Scenario:

1. The administrator clicks the Supporting Staff Management function from the Home Page.
2. The system displays the Supporting Staff Display Page.
3. The administrator chooses to add a new supporting staff record or edit an existing supporting staff record.
4. The system displays the corresponding Add Supporting Staff Page or Modify Supporting Staff Page.
5. The administrator enters or updates the supporting staff information.
6. The system accepts the input.
7. The administrator clicks the Save button.
8. The system validates the input information.
9. The system updates the supporting staff information.
10. The system displays a success message.
11. The use case ends.

Alternative Scenario 1:Empty Input on Add Page

1. The administrator opens the Add Supporting Staff Page.
2. The administrator clicks the Save button without entering the required information.
3. The system detects that the input content is empty.
4. The system displays a warning message.
5. The system remains on the Add Supporting Staff Page.

Alternative Scenario 2: Invalid or Empty Input on Modify Page

The administrator opens the Modify Supporting Staff Page.

The administrator enters invalid information or leaves the required fields empty.

The administrator clicks the Save button.

The system detects that the input content is invalid or empty.

The system displays a warning message.

The system remains on the Modify Supporting Staff Page.

Alternative Scenario 3: Delete Supporting Staff

1. The administrator opens the Supporting Staff Display Page.
2. The administrator selects a supporting staff record.
3. The administrator clicks the Delete button.
4. The system deletes the chosen supporting staff information.
5. The system updates the supporting staff list.
6. The system remains on the Supporting Staff Display Page.

Alternative Scenario 4: Return Without Saving

1. The administrator opens the Add Supporting Staff Page or Modify Supporting Staff Page.
2. The administrator clicks the Back button.
3. The system returns to the Supporting Staff Display Page.

Alternative Scenario 5: Cancel Operation

1. The administrator opens the Supporting Staff Display Page.
2. The administrator clicks the Cancel button.
3. The system cancels the current operation and remains on the Supporting Staff Display Page.

4. External Interface Requirements

4.1 User Interfaces

The user interface (UI) design for the system is provided in a separate document named:

UI_Diagram.pdf

This document contains all UI screens corresponding to the state transition diagrams described in Section 3.

4.2 Hardware Interfaces

The **Mentor Caring System (MCS)** does not require direct control over specialized hardware. It relies on the underlying hardware of client devices and hosting servers to facilitate user interactions and data processing.

4.2.1 Supported Client Devices

Logical Characteristics: The system supports any device capable of running a modern web browser.

Physical Characteristics: Supported devices include desktop computers, laptops, tablets, and smartphones. The physical interfaces involve standard human–machine interface components: keyboards, mice, trackpads, and touchscreens for input, and standard displays for output.

Hardware Communication: Client devices connect to the server via standard physical network interface cards (NICs) supporting Ethernet (IEEE 802.3) or Wi-Fi (IEEE 802.11).

4.2.2 Server Hardware Interfaces

The **Mentor Caring System (MCS)** will be deployed on BNBU's server hardware. The server hardware must provide sufficient memory and CPU resources to handle concurrent database queries and file I/O operations, including parsing Excel files and generating Word documents.

4.2.3 Data and Control Interactions

User inputs are logically captured by the client browser and converted into HTTP/HTTPS requests sent to the server. The server hardware processes these requests and returns HTML/JSON data, which the client hardware renders onto the screen.

4.3 Software Interfaces

The **Mentor Caring System (MCS)** shall interact with external software components required for authentication, notification, and file processing.

4.3.1 School Database Interface

The system shall interface with the BNBU school database for user authentication. The database provides user account, password, name, and role information for staff and students. The system shall send authentication requests during login and receive validation results for access control.

4.3.2 Email Service Interface

The system shall interface with the university email service to send notification emails when a message is delivered in the system. Email notifications shall only inform the receiver that a message exists. Users shall still log in to the system to read and respond to messages.

4.3.3 Excel File Interface

The system shall support Excel file import for the following data:

- faculties, departments, and majors;
- student–mentor allocation data;
- MCP coordinator setup data.

The imported files shall follow the formats defined in the appendices. Imported student allocation data shall include student and mentor information required by the system.

4.3.4 Word File Interface

The system shall support Word file export for student interview records. Exported records shall follow the format defined in the appendix. The exported content shall include interview record details, including interview date and time, problem statement, interview summary, and follow-up actions. Records of different majors shall be exported into different files.

4.4 Communications Interfaces

The **Mentor Caring System (MCS)** shall support communication between client devices, the application server, the school database, and the university email service.

4.4.1 Browser–Server Communication

Users shall access the system through a standard web browser. User operations, including login, search, record maintenance, messaging, appointment handling, import, and export, shall be transmitted between the client and the server through standard web communication mechanisms.

4.4.2 Database Communication

The system shall communicate with the school database during login to validate user identity and obtain role information. The communication shall support authentication for all system users.

4.4.3 Email Notification Communication

The system shall communicate with the university email service to send notification emails after messages are sent in the system. This communication shall support notifications for normal messages and other system messages that require user attention.

4.4.4 File Transfer Communication

The system shall support upload of Excel files from authorized users to the server and download of generated Word files from the server to authorized users. These communications shall support yearly allocation import, MCP coordinator setup import, and interview record export.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Mentor Caring System shall provide acceptable performance for normal operation.

5.1.1 Response Time

- User login authentication shall complete within 3 seconds under normal network conditions.
- Search queries for student, mentor, or log information shall return results within 3 seconds.
- A selected student record or mentoring group member list shall be displayed within 2 seconds.
- Sending an in-system message shall complete within 2 seconds, excluding external email delivery delay.

5.1.2 Import and Export Efficiency

- Import of valid Excel files for student–mentor allocation shall complete within 30 seconds for a normal yearly dataset.
- Import of valid Excel files for MCP coordinator setup shall complete within 30 seconds for a normal departmental dataset.
- Export of student or interview records to a Word file shall complete within 30 seconds for a normal dataset.
- Record filtering for export by academic year, department or major, mentor name, and student name shall complete without noticeable delay.

5.1.3 Concurrent Usage

- The system shall support at least 100 concurrent active users during normal semester operation.
- Operation logs shall be recorded in the background without interrupting normal user operations.
- Normal user operations shall remain available during yearly mentor allocation import.

5.2 Safety Requirements

N/A

5.3 Security Requirements

The system shall provide authentication, authorization, privacy protection, and audit support for student records, interview records, messages, appointments, and operation logs.

5.3.1 Authentication

- All users shall be authenticated through valid university accounts and passwords from the school database.
- Authentication shall be required before accessing protected functions.

- Users shall log in to the system before reading messages, even if email notifications are received.

5.3.2 Authorization

- The system shall enforce role-based access control.
- Faculty consultants shall access only information related to their own faculty.
- Mentors shall access only records of students in their own mentoring groups.
- Students shall access only their own mentoring information.
- MCP coordinators shall access only mentor and student information within their own department.
- Supporting staff shall access only functions assigned to supporting staff.

5.3.3 Privacy Protection

- Student records, interview records, message records, and log information shall be protected from unauthorized access.
- Email notifications shall not disclose sensitive message content.
- Only authorized users shall perform Excel import and Word export operations.

5.3.4 Auditability and Session Security

- The system shall record login time, logout time, and major user operations for audit purposes.
- Log information shall be protected from unauthorized modification or deletion.
- User sessions shall be invalidated after logout.

5.4 Software Quality Attributes

The system shall emphasize adaptability, reliability, usability, maintainability, correctness, and interoperability.

5.4.1 Adaptability

- The system shall support the addition of new faculties, departments, and majors without program code modification.
- Organizational structure information shall be configurable manually or through Excel import.
- Annual mentor allocation updates shall preserve existing student information.

5.4.2 Reliability

- Historical student information shall remain preserved after yearly mentor allocation import.
- Student information shall remain accessible after group removal or mentor reassignment.

- The system shall prevent silent data loss during import, export, update, and delete operations.
- The system shall achieve at least 99% availability during normal semester operation, excluding scheduled maintenance.

5.4.3 Usability

- Users shall see only functions appropriate to their roles.
- The system shall provide clear warning and error messages.
- The interface shall use consistent labels and navigation structures.

5.4.4 Maintainability

- The system shall adopt a modular design for user management, record management, messaging, import/export, and logging.
- Configuration data shall be separated from core business logic.
- System logs shall support debugging, auditing, and maintenance.

5.4.5 Correctness

- When duplicated data appear in an imported Excel file, existing valid information shall remain unchanged.
- Exported records shall match the selected academic year, department or major, mentor name, and student name.
- Access restrictions shall be enforced according to user role and organizational relationship.

5.4.6 Interoperability

- The system shall integrate with the school account database for authentication and role identification.
- The system shall support Excel import and Word export in the required formats.
- The system shall interoperate with an email service for message notifications.

6. Other Requirements

6.1 Database requirements

The database shall be designed using a relational model to support the Mentor Caring System.

It shall consist of multiple entities representing users, mentoring groups, records, and communication data, with clearly defined relationships.

The database shall include the following main entities:

- **User**
  Attributes: `userID` (PK), `name`, `email`, `password`, `role`

Description: stores all system users including administrator, faculty consultant, mentor, student, MCP coordinator, and supporting staff.

- **Student**
  Attributes: `studentID` (PK), `major`, `status`
  Relationship: linked to User entity
- **Mentor**
  Attributes: `mentorID` (PK)
  Relationship: linked to User entity
- **Group**
  Attributes: `groupID` (PK), `academicYear`
  Relationship: each group is assigned one mentor and multiple students
- **GroupMembership**
  Attributes: `groupID` (FK), `studentID` (FK)
  Description: represents many-to-many relationship between students and groups over different years
- **InterviewRecord**
  Attributes: `recordID` (PK), `studentID` (FK), `mentorID` (FK), `date`, `content`
- **Appointment**
  Attributes: `appointmentID` (PK), `studentID` (FK), `mentorID` (FK), `timeSlot`, `location`, `status`
- **Message**
  Attributes: `messageID` (PK), `senderID` (FK), `receiverID` (FK), `content`, `timestamp`
- **Log**
  Attributes: `logID` (PK), `userID` (FK), `loginTime`, `logoutTime`, `operation`

The database shall enforce the following relationships:

- One mentor can be assigned to multiple groups.
- One group can contain multiple students.
- One student can belong to different groups in different academic years.
- One mentor can create multiple interview records for different students.
- Messages shall support many-to-many communication between users.

The database shall also support:

- Role-based access control for different users
- Secure storage of user credentials
- Efficient retrieval of user and mentoring data
- Import of group allocation data from Excel files
- Preservation of historical mentoring records across academic years

6.2 Internationalization requirements

The system shall support internationalization to accommodate users from different linguistic backgrounds. English should be provided as the default language of the system interface. The system design should allow future extension to support additional languages if required.

6.3 Legal requirements

The system must comply with relevant institutional regulations regarding data privacy and information protection. Personal information of users, including student records and mentor information, must be protected and accessible only to authorized users.

6.4 Project reuse objective

The system should be designed with modular components so that certain modules (such as the user authentication module and database access module) can be reused in future university software systems if needed.

Appendix A: Glossary

MCP: Mentor Caring Program.
Faculty Consultant: Departmental advisor, responsible for designating the MCP coordinator and the import list.
MCP Coordinator: Departmental Coordinator, responsible for handling special cases forwarded by mentors.

Appendix B: Analysis Models

This appendix presents the analysis models of the Mentoring and Consultation System.
Context model is presented in Section 2.1 of this report.
Use case diagram is presented in Section 2.2 of this report.

B.1 Class Diagram

The class diagram of the Mentoring and Consultation System is shown in Figure 22. Only the participating classes are included. The important attributes and operations needed to understand the system structure are presented in the diagram.

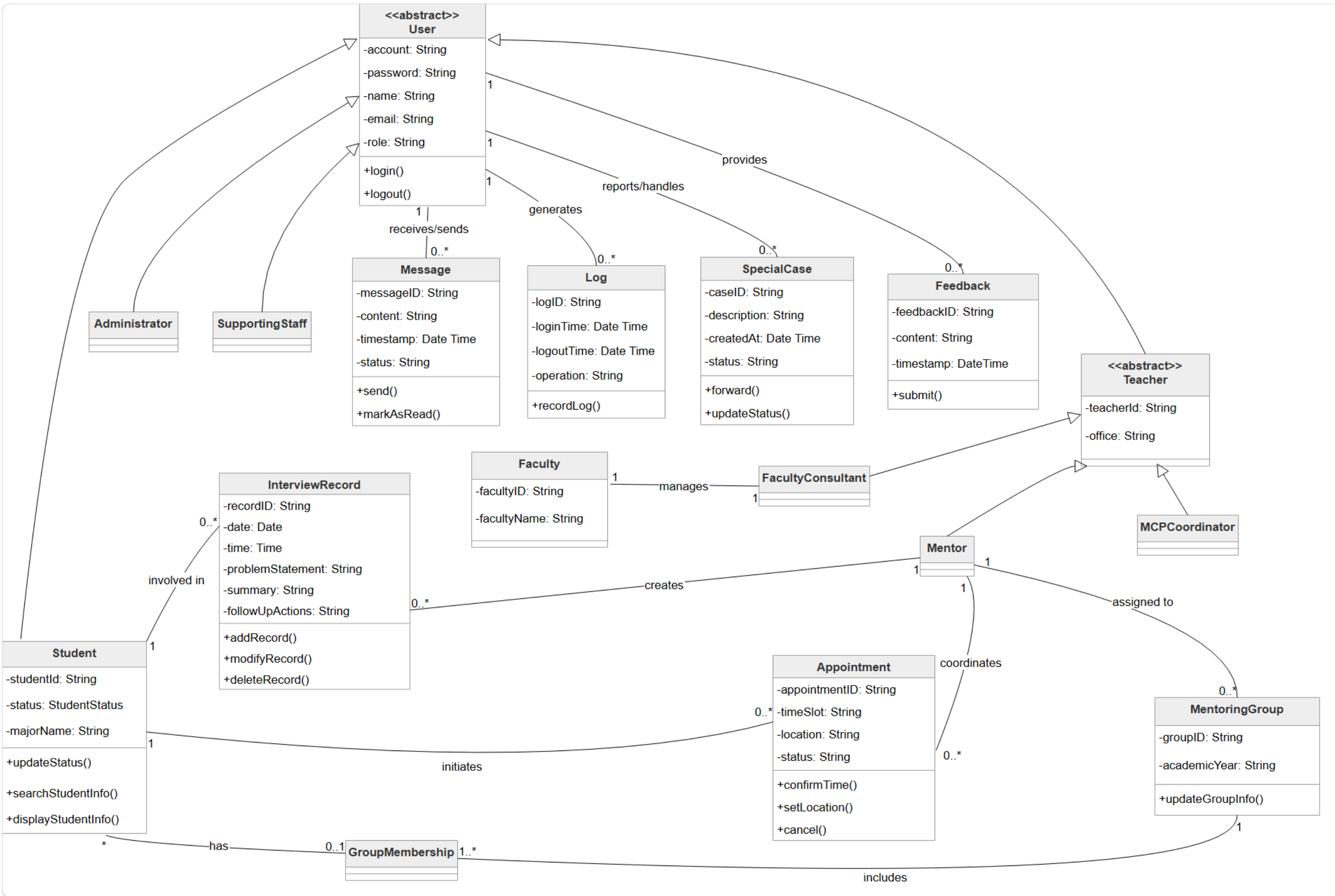

Figure 22: Class Diagram of the Mentoring and Consultation System

B.2 Sequence Diagram

The sequence diagram for the mentor's use case *Search students' info* is shown in Figure 23. It illustrates the interaction among the mentor, the user interface, the control object, and the database when searching for student information.

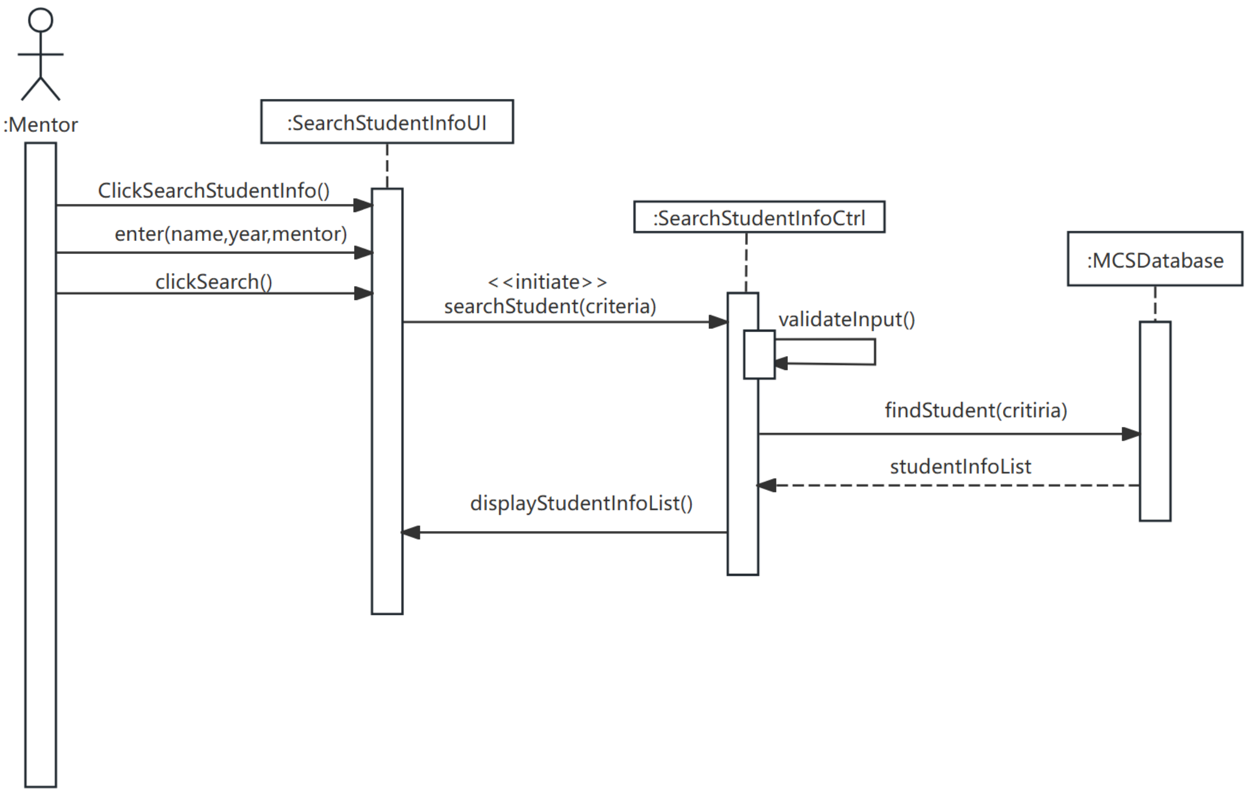

Figure 23: Sequence Diagram for the Use Case "Search Students' Info"

Appendix C: Issues List

The following issues remain to be clarified during later stages of the system development process:

- Final confirmation of database schema
- Detailed access permissions for each user role
- Possible additional system features for future versions
- Final design of the user interface

These issues will be reviewed and resolved during the system design phase.